MIL-T-46386A(MU)
15 September 1967
SUPERSEDING
MIL-T-46386(MU)
1 August 1963

MILITARY SPECIFICATION

TELESCOPE, PANORAMIC: M113

1. SCOPE

1.1 This specification covers one type of telescope known as Telescope, Panoramic: M113 primarily intended for laying the 105MM Howitzer M102 in azimuth for indirect fire.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-O-13830	Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly and Inspection of
MIL-F-13926	Fire Control Materiel; General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel, Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

DRAWINGS

U.S. Army, Frankford Arsenal

F8626189	Telescope, Panoramic: M113

PACKAGING DATA SHEET

MIL-P-14232/P8626189	Packaging of Telescope, Panoramic: M113

FSC 1240

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(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.- Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

3.2 Fabrication.- The telescope shall be manufactured in accordance with Drawing F8626189, "Telescope, Panoramic: M113" and drawings pertaining thereto.

3.3 General specifications.- The contractor shall be responsible for adherence to and compliance with the following requirements of MIL-0-13830 and MIL-F-13926.

3.3.1 MIL-0-13830.

(a) Cleanliness.
(b) Illuminated reticle.

3.3.2 MIL-F-13926.

(a) Order of precedence.
(b) Dimensions and tolerances.
(c) Inorganic protective surface finishes.
(d) Part identification and marking.
(e) Electrical and electronic assemblies.
(f) Workmanship.

3.4 Environmental.

3.4.1 Storage temperature.- The telescope shall show no evidence of physical failure and shall meet all requirements of this specification at standard ambient temperature (plus 60 degrees to 90 degrees Fahrenheit) after having been exposed and thermally stabilized at ambient temperatures of plus 160 degrees and minus 80 degrees Fahrenheit.

3.4.2 Operating temperature.- The telescope shall meet the applicable requirements of 3.9 while exposed and thermally stabilized at ambient temperatures of plus 150 degrees and minus 40 degrees Fahrenheit. Upon return to standard ambient temperature, the telescope shall meet the requirements of 3.4.3.

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3.4.3 Vibration.- The telescope shall be vibrated in a vertical plane at a constant frequency of thirty cycles per second with an amplitude of 1/16 inch (1/8 inch total excursion) for a period of five minutes plus or minus fifteen seconds. Subsequent to vibration there shall be no evidence of physical failure and the telescope shall meet the requirements of 3.5 to 3.12 inclusive.

3.5 Cleanliness.- The telescope shall meet the cleanliness requirements of MIL-O-13830 and there shall be no more than 2 particles of dirt visible on the reticle. The area of each particle shall not be in excess of the areas equivalent to a size 10 dig. There shall be no dirt on other surfaces which interferes with observation or adversely affects image quality.

3.6 Sealing.

3.6.1 Pressure (optical assembly).- The interior atmosphere of the optical assembly shall be capable of withstanding an internal pressure of 5 ± 0.10 pounds per square inch gage (psig) for a period of 5 minutes with no evidence of leakage.

3.6.2 Pressure (counterbox).- The counterbox shall be capable of withstanding 2.0 + 0.20 psig for a period of 3 minutes with no evidence of leakage.

3.7 Accuracy.

3.7.1 Collimation.- With the flat mounting surface of the telescope in a horizontal plane the azimuth counter reading 3200 mils, the reset counter shall read 3200 mils when the left (L) and right (R) gunner's aid counters are set at 00 and 00. The eyepiece elbow shall be positioned in detent and shall be perpendicular to the plane of the telescope locating keyways. With the head cover open the forward line of sight shall be to the left of the eyepiece elbow and shall be horizontal and parallel to the plane of the telescope locating keyways.

3.7.1.1 Head cover closed.- With the telescope positioned as in 3.7.1, the head cover in the closed position, the line of sight shall be horizontal through the hole in the cover and shall be coincident with the geometric axis of the reticle.

3.7.1.2 Light projector.- With the telescope positioned as in 3.7.1, the light projector lamp energized, the vertical slit in the head of the telescope shall be illuminated and shall be clearly visible over its entire length and width at a target point from a distance of 80 meters.

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3.7.1.3 Open sight.- With the telescope positioned as in 3.7.1, the line of sight of the open sight shall be horizontal within plus or minus 10 mils and parallel in deflection (azimuth) to the optical line of sight within plus or minus 10 mils.

3.7.1.4 Eyepiece elbow movenment.- Movement of the eyepiece elbow from its midpoint to the left and right extremes of travel shall not cause a deviation in excess of one degree.

3.7.2 Image tilt.- The image of a vertical target line shall be vertical within one degree throughout one revolution of the head.

3.7.2.1 Image wander.- The wander of the reticle image measured at the eyepiece shall not exceed 4 degrees in diameter for 2 full revolutions of the head.

3.7.3 Plumb travel.- With the telescope positioned as in 3.7.1, the line of sight shall follow a plumb line (starting with the line of sight horizontal and in coincidence with the plumb line) within 0.5 mil when the head prism is depressed a minimum of 300 mils and elevated 300 mils from the horizontal when checked at any deflection setting.

3.7.4 Parallelism of reticle and image.- With the telescope positioned as in 3.7.1, the vertical reticle line shall be parallel to the image of a plumb line wihtin 30 minutes of arc at any deflection setting.

3.7.5 Level travel.- With the telescope positioned as in 3.7.1, the line of sight shall not deviate from a horizontal target line by more than 1.0 mil (total excursion of 2 mils) when the head is rotated through two revolutions.

3.7.5.1 Lift.- Vertical displacement of the line of sight due to reversing direction of rotation of the head shall not exceed 1.0 mil.

3.7.6 Deflection error.

3.7.6.1 Throughout 800 mil steps.- Error at deflection readings in 800 mil steps throughout two full revolution of the head in one direction shall mot exceed 1,0 mil including backlash. The total spread between the greatest positive error and the greatest negative error shall not exceed 1.0 mil. The total error, excluding backlash, between the 6400 mil azimuth counter reading and the reset counter for any one number shall not exceed 0.25 mil.

3.7.6.2 Throughout 15 mil steps.-Error at deflection readings in 15 mil steps taken at eight random positions throughout one full revolution of the head shall not exceed 0.2 mil excluding backlash at each position.

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3.7.6.3 Backlash.- Backlash in the deflection mechanism shall not exceed 0.5 mil when read on the 6400 mil azimuth counter and 0.75 mil when read on the reset counter. Backlash in the elevating mechanism shall not exceed 1.0 mil.

3.7.7 Eyepiece focus.- Eyepiece focus of the horizontal reticle line and vertical reticle line at the center of the field shall each be within the limits of 0.75 and 1.0 diopters.

3.7.8 Resolution.- Resolution on the optical axis shall be 15 seconds of arc or less when using an auxiliary telescope of at least 3 power having an entrance pupil of at least 0.4 inch. Resolution shall consist of detectable line structure and proper line count in all 4 meridians. Variation in focus between meridians shall not exceed 0.25 diopters as measured at the panoramic eyepiece.

3.7.9 Parallax.- Parallax in the center of the field shall not exceed 0.35 mil when viewing a target between 40 meters and infinity. (Parallax is nominally zero at 80 meters.)

3.7.10 Reticle accuracy.- The total angular subtense of the reticle graduations in object space shall be as indicated on the reticle drawing within two percent.

3.8 Counter operation.

3.8.1 Reset counter.- The reset counter shall be resettable to 3200 mils from any counter reading. The reset counter and the 6400 mil azimuth counter shall follow when the operating knob is turned at the rate of 200 revolutions per minute (RPM) for a period of one minute. The gunner's aid counter shall remain at R-00 and L-00.

3.8.2 Gunner's aid counter.- The stops shall allow an excursion of plus 50 mils and minus 50 mils.

3.8.2.1 Counter setting.- Setting right (R) 50 mils on the gunner's aid counter shall increase the reset counter by 50 mils. Setting left (L) 50 mils on the gunner's aid counter shall decrease the reset counter by 50 mils. No change shall result in the 6400 mil azimuth counter.

3.8.2.2 Direct fire, click lead, engagement knob.- With the detent on the knob of the 6400 mil azimuth counter set for direct fire the azimuth counter knob shall indicate a distinguishable click, indicating detent position at each five mil interval between 0 and 6400 mils.

3.9 Operability of movable parts.

3.9.1 Torque.- The running torque required to rotate the following components shall be within the values tabulated. When checked at the

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temperature specified in 3.4.2, the running torque values shall not exceed the value obtained at standard ambient temperature by more than 100%. 150%

Elbow Assembly	~~7~~ to 12 inch pounds
Elevation Knob	~~10~~ 5 to 48 inch ounces
Reset Counter (rotation to reset counter)	2 to 40 inch ounces
Azimuth Knob	4 to 10 inch pounds
Boresight Adjustment (rotation while clutch is disengaged)	2 to 20 inch ounces

3.9.2 Operability.- This requirement shall be met with the telescope thermally stabilized at the temperatures specified in 3.4.2. The reset counter knob shall be manually operable to push in for resetting counters (direct force). The boresight adjustment shall be operable to disengage the clutch for boresighting (direct force). The azimuth counter cover and latch, head cover and latch, azimuth knob folding handle, elbow detent lock pin and spring action of the electrical contact shall be operable.

3.10 Illumination.

3.10.1 Reticle.- When observed in a darkened area, with reticle illuminated, the reticle markings shall be clearly defined throughout one revolution of the telescope head.

3.10.2 Counters.- Counter illumination shall be sufficient for the numerals and markings of each counter to be clearly distinguishable when observed in a darkened area.

3.11 Counter numerals.- With the lowest digit of each counter in coincidence with its respective index line, the remaining numbers shall be in horizontal alignment within one-sixteenth inch when checked at the standard ambient temperature.

3.12 Electrical.

3.12.1 Insulation resistance.- Insulation resistance shall not be less than 10 megohms with a potential of 100 volts direct current applied.

3.12.2 Continuity.- The continuity of the wiring shall be checked for conformance with the applicable wiring and continuity drawing requirements.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise

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specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production)submission.-When required by the contract, an initial production sample shall be submitted to a Government approved facility for test and evaluation in accordance with the contract schedule. The sample shall consist of three telescopes. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply. After inspection and provisional acceptance at source, the initial production sample shall be inspected at a Government approved facility for all requirements of the drawings, and specifications. The Government reserves the right to require new initial production samples until such time as an acceptable sample is submitted.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the List of Inspection Documents (IEL), when provided with the technical data package (TDP). In the absence of SQAP's the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

4.3.2.2 Final acceptance inspection.- Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Table I herein. Examination and tests for Packaging, Packing and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Table I shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

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TABLE I - CLASSIFICATION OF DEFECTS

CLASS	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
CRITICAL: 100% INSPECTION			
1.	Pressure (Optical Assy)	3.6.1	4.6.1.1
2.	Pressure (Counter Assy)	3.6.2	4.6.1.1.1
MAJOR: AQL 1.0 Defective			
101.	Previbration collimation	3.7.1	4.7.2.1
102.	Vibration	3.4.3	4.7.2.1.1
103.	Post-vibration collimation	3.4.3	4.7.2.1.2
104.	Image wander	3.7.2.1	4.7.2.1.3
105.	Cleanliness	3.5	4.7.2.1.4
106.	Image tilt	3.7.2	4.7.2.1.5
107.	Parallelism of image & reticle	3.7.4	4.7.2.1.6
108.	Eyepiece focus	3.7.7	4.7.2.1.7
109.	Parallax	3.7.9	4.7.2.1.8
110.	Reticle accuracy	3.7.10	4.7.2.1.9
111.	Settings, counter, gunner's aid	3.8.2.1	4.7.2.2.1
112.	Counter stops	3.8.2	4.7.2.2.1
113.	Deflection error (800 mil)	3.7.6.1	4.7.2.2.2
114.	Deflection error (15 mil)	3.7.6.2	4.7.2.2.2
115.	Azimuth backlash	3.7.6.3	4.7.2.2.2
116.	Level travel	3.7.5	4.7.2.2.3
117.	Lift	3.7.5.1	4.7.2.2.4
118.	Click mechanish	3.8.2.2	4.7.2.2.5
119.	Rapid traverse (reset Counter)	3.8.1	4.7.2.2.6
120.	Resolution	3.7.8	4.7.2.2.7
121.	Plumb travel	3.7.3	4.7.2.3.1
122.	Torque (60°-90°F)	3.9.1	4.7.2.4.1
123.	Operability (60°-90°F)	3.9.2	4.7.2.4.1
124.	Illumination (reticle)	3.10.1	4.7.2.5
125.	Illumination (counters)	3.10.2	4.7.2.5
126.	Light projector	3.7.1.2	4.7.2.5
127.	Counter numbers	3.11	4.7.2.6
128.	Elbow movement	3.7.1.4	4.7.2.1.6
129.	Insulation resistance	3.12.1	4.7.2.7
130.	Continuity	3.12.2	4.7.2.8
MINOR: AQL 2.5 Defective			
201.	Head cover	3.7.1.1	4.7.2.1.2
202.	Open sight	3.7.1.3	4.7.2.2.8
203.	Excursion range (elevation)	3.7.3	4.7.2.3.1
204.	Elevation backlash	3.7.6.3	4.7.2.3.2

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

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4.4 Special sampling for environmental testing.

4.4.1 Samples shall be selected from lots at random using inspection Level S-3 Table I of MIL-STD-105. Acceptance or rejection of the lot shall be in accordance with AQL 2.5 of Table II-A of MIL-STD-105. The samples shall have met all the requirements and tests outlined in Table I prior to being subjected to the following examination and test.

TABLE II

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301. Fabrication	3.1	Certification
302. General specification	3.2	Certification
303. Temperature storage	3.4.1	4.6.1.2
304. Temperature operating	3.4.2	4.6.1.2
305. Operability (storage & oper. temp)	3.9.2	4.7.2.4
306. Torque (storage & oper. temp)	3.9.1	4.7.2.4.1

4.4.2 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the List of Inspection Documents when provided with the Technical Data Package (TDP).

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerance used in the design of test equipment

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shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.3 Test equipment.- In conjunction with 4.5.2.2 the following standard test equipment shall be utilized in the performance of the applicable tests as specified in 4.6 and 4.7.

NOMENCLATURE	DESCRIPTION
1. Hot & cold chamber	Standard type conforming to the accuracies outlined in test facilities specified under test facilities of MIL-F-13926
2. Continuity checker	Standard shop type capable of providing readings accurately within 2% of specified tolerance
3. Insulation resistance checker	Standard shop type conforming to the accuracies outlined in 4.5.2.2
4. Power supply	0 to 100V D. C. output within 10% accuracy
5. Torque wrench and scale	Standard shop type accurate to within 10% of drawing tolerance

4.6 Test methods and procedures.

4.6.1 Environmental.

4.6.1.1 Temperature test.-The panoramic telescope shall be subjected to test Procedure I of MIL-F-13926; however, the temperatures specified in 3.4.1 and 3.4.2 shall apply. When the panoramic telescope has been stabilized at the temperature specified, the result of the specified test shall conform to the limits outlined in 3.4.1 and 3.4.2 using test procedures outlined herein.

4.6.1.2 Sealing optical assembly.- This test shall be performed using testing equipment the accuracy of which shall conform to the accuracy depicted on Drawing F8565556. When using pressure test fixture, the fixture shall be set-up as outlined in set-up instructions shown on Drawing F8565556,

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Sheet 1. The gas used shall be dry nitrogen having a dew point of -25°F. After connecting the test fixture to the panoramic telescope, adjust testing pressure to the pressure specified in 3.6.1 and maintain the pressure for time specified in 3.6.1. Pressure gage shall remain constant to assure compliance with 3.6.1. At the completion of this test connect a dew point tester to the exit post and purge the optical assembly until -25°F is recorded and seal the unit.

4.6.1.2.1 Sealing (counter assembly).- This test to be accomplished as outlined above, however, the pressure and time values specified in 3.6.2 shall apply.

4.7 Performance testing.

4.7.1 Orientation.- Proper orientation of the panoramic telescope is assured when mounted and secured to an adapter, the accuracy of which shall be equal to or better than the accuracy depicted on Drawing F8565945, and when aligned as outlined in set-up instructions specified on Drawing F8213899. With the panoramic telescope positioned on the fixture as outlined above the line of sight, elbow assembly, counters and the five mil click mechanism shall be set as specified in 3.7.1.

4.7.2 Collimation.

4.7.2.1 Previbration.- With telescope positioned as outlined in 4.7.1 observe coincidence between the panoramic telescope line of sight and the optical axis of the test reticle (target) in the collimator projector. This coincidence must be within the limits outlined in 3.7.1. Record this position. Exercise care not to move the setting of the panoramic telescope.

4.7.2.1.1 Vibration.- This test shall be performed using a testing device, the accuracy of which shall be equal to or exceed the accuracy depicted on Fixture Drawing F7560085. The test shall be performed as outlined in Procedure V of MIL-F-13926; however, the duration shall be as specified in 3.4.3. At the conclusion of the test, the telescope shall be subjected to a visual and tactile examination and shall be subjected to the following performance test to determine conformance with 3.4.3.

4.7.2.1.2 Post vibration.- Immediately after completing the vibration test reassemble the telescope to the adapter and without disturbing any previous settings observe where the panoramic telescope's optical line of sight falls upon the projector collimator reticle. This observed point and the recorded position (4.7.2.1) shall coincide within the limits specified in 3.4.3. Close the cover and observe where the line of sight falls with respect to the recorded position of the optical axis. They must coincide within the limits outlined in 3.7.1.1.

4.7.2.1.3 Image wander.- With the cover opened and with the telescope positioned as outlined in 4.7.1, rotate the telescope azimuth mechanism in one direction to any azimuth setting and rotate the azimuth mechanism of the test fixture and equal amount in the opposite direction. Now observe coincidence between the optical axes. Non-coincidence between the axes shall not exceed an 18 mil spread to assure compliance with 3.7.2.1.

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This spread to be measured on the panoramic telescope reticle.

4.7.2.1.4 Cleanliness.- The panoramic telescope shall be tested for cleanliness by using the shadowing techique outlined in MIL-O-13830. At the conclusion of this test, dirt particles shall not exceed the limits outlined in 3.5.

4.7.2.1.5 Image tilt.- With the panoramic telescope and fixture positioned as outlined in 4.7.1, image tilt shall be observed by sighting through Collimator F5549108 aligned to the optical axis of the eyepiece assembly and note parallelism between its vertical reticle line and the vertical line of the collimator target. Both lines shall be parallel within the limits provided on the projected target (collimator reticle) to assure conformance with the 3.7.2. Repeat this test with the elbow assembly against each of the stops to assure conformance with 3.7.1.4. Again observe parallelism between the reticle lines. Repeat this test at eight different azimuth counter settings in 6400 mils of rotation. Each setting in azimuth will require the azimuth fixture plate to be rotated an equal amount in an opposite direction to permit reestablishing the line of sight to the position observed in 4.7.2.1.2. All readings are to be within the limits outlined in 3.7.1.4.

4.7.2.1.6 Parallelism of image and reticle.- With the panoramic telescope positioned as outlined in 4.7.1, align one end of the instruments vertical reticle line to the projector collimator's vertical reticle line and observe where the opposite end of the panoramic telescope verticle reticle line falls with respect to the projector collimator vertical line, both lines must coincide within the limits specified in 3.7.4. Repeat this procedure with elbow assembly against each of the extreme stops. Also repeat this procedure at approximately eight different azimuth settings 6400 mils of rotation. Return the line of sight to the projector collimator's target line after each selected azimuth setting by rotating the azimuth fixture plate an equal amount in the opposite direction. All check points are to be within the limits outlined in 3.7.1.4 and 3.7.4.

4.7.2.1.7 Eyepiece focus.- This test is to be accomplished by placing a white background in front of the panoramic telescope window. Eyepiece focus is to be measured using a 3 power dioptometer conforming to the accuracies depicted on Dwg 7680631. Adjust the dioptometer eyepiece assembly to the dioptometer's reticle until a clear sharp image appears on the dioptometers optical axis. Next direct the dioptometer's line of sight toward a 500 feet distant tartet such as a smoke stack, edge of a building, or some other well defined target image. Adjust the objective assembly until parallax is removed. Parallax is defined as any apparent movement between the reticle and target image when the head is moved from side to side or up and down. Align the dioptometer's optical axis to the panoramic telescope optical axis and obtain a clear sharp image of the panoramic telescope reticle on the optical axis. Make all measurements on the dioptometer by adjusting the objective assembly. Any and all adjustments to be within the limits of 3.7.7.

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4.7.2.1.8 Parallax.- Using the set-up outlined in 4.7.1, parallax will be checked after each of the distances specified in 3.7.9 has been set into the projector collimator. The test is performed by aligning a reticle line or mark near or on the optical axis the projector collimator reticle. Observe this positioning, there shall be no apparent movement between the selected lines or mark when the observer moves his head from side to side or up and down. Reticle displacement shall not exceed the limits outlined in 3.7.9. Use the thickness of the reticle line to measure apparent movement.

4.7.2.1.9 Reticle accuracy.- Reticle accuracy may be assured by aligning the intersecting point of the panoramic telescopes verticle and horizontal reticle line to the corresponding intersecting reticle lines of the projector collimator. Adjustments may be made by means of the panoramic telescopes azimuth and elevating mechanisms. When coincidence has been established, observe where the 80 mil graduations of the panoramic telescope's horizontal reticle line falls with respect to the corresponding reticle line of the projector collimator. They must coincide within the limits of the reticle tolerance mark to assure compliance with 3.7.10.

4.7.2.2 Counter operation.

4.7.2.2.1 Gunner aid counters.- This test may be accomplished with the panoramic telescope positioned as outlined in 4.7.1 with the line of sight falling upon the projector collimators' reticle as outlined in 4.7.2.1.2. Observe and record the azimuth test fixture reading. Set any correction reading into either the "L" or "R" counter. Any reading induced into the right "R" gunner's aid counter shall increase the reset counter and equal amount and any reading induced into the left "L" gunner's aid counter shall decrease the reset counter and equal amount to assure compliance with 3.8.2.1. When 50 mils has been inserted into either gunner's aid counter the reset counter shall reflect an increase or decrease of 50 mils within the limits outlined in 3.8.2.1 and the correction counter shall have traveled the minimum excursion outlined in 3.8.2 and the 6400 mil counter shall remain unchanged (see 3.8.2.1). The test for accuracy shall be performed by inserting a correction into either the "L" or "R" gunner's aid counter and observing the appropriate change in the reset counter. Turn the azimuth fixture plate to traverse the line of sight in a clockwise direction an equal amount when a correction is placed into the "L" counter. By means of the azimuth knob of the panoramic telescope traverse the line of sight in the opposite direction until the reset counter records 3200 mils. Now sighting through the panoramic telescope and observe coincidence between the panoramic telescopes optical axis and the point on the projector collimators reticle established in 4.7.2.1.2. This coincidence must be maintained within the limits specified in 3.8.2.1. Repeat this procedure at two additional correction readings and at the 50 mil correction setting of each gunner's aid counter.

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4.7.2.2.2 Deflection error, backlash and 15 mil settings.- With the panoramic telescope positioned as outlined in 4.7.1 rotate the azimuth fixture plate 800mils. Sight through the panoramic telescope and rotate the azimuth knob in the opposite direction until the line of sight falls in coincidence with the position outlined in 4.7.2.1.2. Do not override this point of coincidence. Record deflection counter reading and reset counter reading. Both should be in conformance with 3.7.6.1. Rotate the deflection knob one complete revolution and retrun the line of sight to the previously established position and observe deflection and reset counter readings. This difference, if any, is backlash and shall not exceed the limits of 3.7.6.3. Rotate the azimuth plate an additional 15 mils and repeat above procedure. When coincidence is obtained observe deflection and reset counter readings. Coincidence must conform to the limits outlined in 3.6.7.2. Repeat the 15 mil check after each 800 mil setting.

4.7.2.2.3 Level travel.- This test may be performed in conjunction with the deflection test outlined in 4.7.2.2.2. The test is performed by sighting through the panoramic telescope after a selected azimuth setting has been inserted into the fixture and panoramic telescope, observe coincidence between the telescope horizontal reticle line and the horizontal target line near the optical axis. They must coincide within the limits outlined in 3.7.5.

4.7.2.2.4 Lift.- This test may be performed in conjunction with the backlash test outlined in 4.7.2.2.2. The test is performed after returning the line of sight to the point of coincidence by observing where the horizontal reticle line of the panoramic telescope falls with respect to the previously established position (4.7.2.2.3). They must coincide within the limits of 3.7.5.1.

4.7.2.2.5 Five mil click mechanism.- This test may be performed by using visual and tactile inspection techniques. The result of this inspection shall conform to the requirement outlined in 3.8.2.7 when performed at the temperature specified.

4.7.2.2.6 Rapid traverse (reset counter).-This test is made by assembling a testing mechanism capable of providing an RPM output specified in 3.8.1 to the azimuth knob of the panoramic telescope by means of an appropriate adapter. After subjecting the panoramic telescope to the RPM for the time specified, the reset counter and the azimuth counter shall follow as required in 3.8.1 and the reset counter shall reset to 3200 mils at the end of each speed test required in 3.8.1. The gunner's aid counter shall remain as specified in 3.8.1.

4.7.2.2.7 Resolution.- This test may be performed after directing the line of sight toward a target similar to that specified in MIL-O-13830. The actual test is performed as outlined in MIL-O-13830. Observe the definition pattern on the target. The patterns must be resolved in the four meridians within the limits outlined in 3.7.8.

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4.7.2.2.8 Open sight.- This test is accomplished by using the set-up in 4.7.1. The test is performed by positioning the optical axis of the auxiliary telescope, F5549108 close to, but not on the vertex of the "V" grooves of the open sight and observing where the optical axis of the auxiliary telescopes fall upon the target with respect to the 20 mil square of the target, see Sheet 8 of Drawing F8213899. The optical axis must fall within the square to conform with the tolerance specified in 3.7.1.3.

4.7.2.3 Elevation.

4.7.2.3.1 Elevation and plumb travel.- Return the panoramic telescope to the position outlined in 4.7.1. Scribe an index line on the elevation knob in line with an index scribed on the head. Perform these tests by rotating the azimuth fixture plate to direct the panoramic telescope's optical axis toward a plumb line. Position a plumb line in coincidence with the center vertical reticle line. Position the vertical target line in coincidence with the plumb and verticle reticle line and raise or lower the target to obtain coincidence between the target horizontal line and the horizontal reticle line of the panoramic telescope's reticle. The plumb travel test is performed by actuating the panoramic telescopes elevation knob from the zero setting to each extreme. The line of sight must travel to the limit lines of the target to assure compliance with 3.7.3 and must follow the center vertical line from zero elevation within the width of the target limit line to assure compliance with 3.7.3. Repeat this procedure at approximately six different azimuth settings by turning the panoramic telescopes azimuth mechanism in one direction and the rotating surface of the test fixture in the opposite direction to obtain the point of coincidence mentioned above.

4.7.2.3.2 Elevation backlash.- This test can be performed in conjunction with the previous test by sighting through panoramic telescope and by means of the elevation knob raise or lower the line of sight to a selected line on the target (do not override this target line) and scribe a line on the head casting in line with the index on the elevation knob. Rotate the elevation knob past this selected target line and return to the coincided lines mentioned and observe where the optical axis (line of sight) falls with respect to the previously selected target line. This difference, if any, as measured by the thickness of the reticle line, shall not exceed the limits outlined in 3.7.6.3.

4.7.2.4 Operability.- When the panoramic telescope has been stabilized at the extreme operating temperatures specified in 3.9.2, tactile inspection techniques shall be used to determine compliance outlined in 3.9.2.

4.7.2.4.1 Torque.- Torque tests shall be performed under temperature conditions outlined in 3.9.1. The torque tests shall be performed using a torque wrench and appropriate adapter. The torque readings obtained shall conform to the limits specified in 3.9.1.

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4.7.2.5 Illumination.- Electrically connect a power source to the panoramic telescope capable of producing the voltage specified and energize the appropriate lamps and visually inspect to determine compliance with 3.10.1 and 3.10.2. Rotation of the rheostat knob from the off position shall increase intensity of illumination from dim to bright and the projector light shall be visible within the limits outlined in 3.7.1.2.

4.7.2.6 Counter numbers.- By means of standard measuring equipment determine digit alignment in accordance with 3.11.

4.7.2.7 Insulation resistance.- This test shall be performed as outlined in Test Method 302 of MIL-STD-202, using test condition "A". The test shall be made between insulated points and ground. Resistance shall not be less than the value specified in 3.12.

4.7.2.8 Continuity.- Use standard shop type continuity checks in conjunction with standard inspection techniques.

5. PREPARATION AND DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with MIL-P-14232, P8626189. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The Panoramic Telescope: M113 when mounted on Telescope Mount: M134, is primarily intended to lay the 105MM Howitzer M102 in azimuth for indirect fire.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable levels of preservation, packaging and packing.
(c) Applicable stock number.

Custodian:

Army - MU

Preparing activity:

Army - MU

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